

AI SEO Agent - 2026 Edition 🚀

> From Optimization to Authority: The New Frontier of Search

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📋 Overview

The **AI SEO Agent** is a cutting-edge command-line tool designed to audit and optimize websites for the Generative AI era. Unlike traditional SEO tools that focus solely on keywords and backlinks, this agent evaluates your digital presence against **Generative Engine Optimization (GEO)** standards, **Search Everywhere** strategies, and Google's **E-E-A-T** (Experience, Expertise, Authoritativeness, Trustworthiness) guidelines.

Its mission is simple: **To help your content get cited by AI, not just ranked by algorithms.**

✨ Key Features

- * **🤖 Generative Engine Optimization (GEO)**: Analyzes content structure (definition lists, tables, data) to maximize "Citation Share" in AI-generated answers (ChatGPT, Gemini, Claude).
- * **🌐 Search Everywhere Audit**: Checks for social graph tags and multimodal asset readiness (images, videos) to ensure visibility across all platforms.
- * **🛡️ E-E-A-T & Authority Checks**: Scans for trust signals, author bylines, and expert validation markers.
- * **🔧 Technical SEO Foundation**: Validates critical basics like meta tags, canonicals, and schema markup presence.
- * **📊 Agentic Assets**: Automatically generates modern assets like ``suggested_robots.txt`` (for AI bot permissions) and ``suggested_schema.json``.
- * **💻 Rich CLI Interface**: Features a beautiful, interactive terminal UI with live progress tracking and color-coded summaries.

🛠️ Installation

1. ****Clone the Repository****

```
```bash
git clone https://github.com/sgins/SG-SEO.git
cd SG-SEO
```
```

2. ****Set up the Environment****

It is recommended to use a virtual environment.

```
```bash
Windows
python -m venv venv
.\venv\Scripts\activate

Mac/Linux
python3 -m venv venv
source venv/bin/activate
```
```

3. ****Install Dependencies****

```
```bash
pip install -r requirements.txt
```
```

🚀 Usage

The agent is run via the command line. The main entry point is ``src/main.py``.

Quick Start

Analyze a website (defaults to single page depth):

```
```powershell
python src/main.py https://www.your-website.com/
```

```

Advanced Usage

| Option | Description | Default |
|-----------------------------|--------------------------------------|--------------------------|
| Option | Description | Default |
| Option | Description | Default |
| <code>`url`</code> | The target URL to analyze (required) | N/A |
| <code>`--depth`</code> | How deep to crawl internal links | <code>`1`</code> |
| <code>`--output-dir`</code> | Directory to save generated reports | <code>`./reports`</code> |

****Example: Deep Crawl****

To analyze the homepage and all directly linked pages:

```
```powershell
python src/main.py https://www.your-website.com/ --depth 2
```
```

📁 Output & Reports

After a successful run, the agent creates a timestamped report folder (default: ``reports/``).

1. The Audit Report (``SEO_Report_YYYYMMDD.md``)

A comprehensive Markdown file containing:

- * ****Executive Summary****: High-level health check.
- * ****Page-by-Page Analysis****: detailed breakdown of Technical, GEO, and E-E-A-T scores.
- * ****Strategic Recommendations****: Actionable steps to improve AI visibility.

2. Suggested Assets

- * ****`suggested\_robots.txt`****: A modernize robots file that explicitly allows helpful AI bots (like GPTBot) to crawl your site, increasing your chances of being cited.
- * ****`suggested\_schema.json`****: A template JSON-LD structure to help search

engines understand your entity (Organization, Website).

🧠 Under the Hood

The project is structured for modularity and extensibility:

- * ``src/main.py``: CLI entry point and UI orchestration.
- * ``src/crawler.py``: Polite web crawler with depth control.
- * ``src/analyzer.py``: The brain of the operation. Contains logic for GEO scoring, E-E-A-T detection, and technical checks.
- * ``src/reporter.py``: Generates the human-readable reports and machine-readable assets.

📖 User Manual: The Educational Guide to Implementation

This section provides comprehensive instructions for executing the SEO and Google Analytics (GA4) strategies across different hosting environments. Before we dive into the commands and clicks, let's establish a shared understanding of the core concepts you will encounter.

📚 Glossary of Core Concepts

- * **SEO (Search Engine Optimization)**: The practice of improving a website's technical configuration, content relevance, and link popularity so search engines rank it higher.
- * **Google Analytics 4 (GA4)**: The latest version of Google's analytics platform, focusing on event-based tracking rather than pageviews, allowing you to understand user journeys across websites and apps.
- * **Google Cloud & "Vibe Coding"**: A custom hosting environment where you have full control over the server and file architecture. "Vibe Coding" refers to writing minimal, effective scripts (like PowerShell) to automate complex tasks rather than relying on bloated software.
- * **Wix (Managed CMS)**: A Content Management System where the underlying servers and code are managed for you. You interact with it through a dashboard rather than writing raw code.
- * **Sitemap (XML Sitemap)**: A file that lists all the important pages on your website. It acts as a roadmap for search engine crawlers (like Googlebot) so they

know exactly what content exists and when it was last updated.

* **Robots.txt:** A text file placed at the root of a website that gives instructions to web robots (crawlers) about which pages they are allowed to index. "AI-Optimized" means explicitly allowing AI bots (like GPTBot) to read your site.

* **Title Tags & Meta Descriptions:** HTML elements in a page's `<head>`. The Title is the clickable blue link in search results, and the Meta Description is the short summary below it. Both are critical for encouraging users to click.

* **Canonical Tags:** An HTML element that tells search engines which version of a URL is the "master" copy. This prevents penalties if the exact same content is accessible via multiple URLs (e.g., duplicate content).

* **HREFLANG:** An HTML tag used to specify the language and geographical targeting of a webpage. If you have an English version and a Hebrew (`/he/`) version, this tag tells Google which one to show to which user.

* **Google Tag Manager (GTM):** A tag management system that allows you to quickly update tracking codes and related code fragments (tags) on your website without modifying the raw HTML code yourself.

* **BigQuery:** A fully-managed, serverless data warehouse by Google. You can export GA4 data here to run complex SQL queries or feed it into AI models.

* **LLM (Large Language Model):** AI systems like ChatGPT or your local Ollama instance (e.g., Llama 3) that understand and generate human-like text.

* **Schema Markup (JSON-LD):** Structured data vocabulary that helps search engines better understand the info on your website in order to serve "rich results" (like star ratings, prices, or event dates right in the search results).

* **Google Search Console (GSC):** A free Google tool that helps you monitor, maintain, and troubleshoot your site's presence in Google Search results.

🎓 The Theory & Practice of Modern Digital Presence

Before executing the technical scripts, it is crucial to understand the philosophy driving them. Modern web architecture is no longer just about presenting information to humans; it is about structuring data so that machines—specifically search algorithms and AI models—can ingest, understand, and cite your expertise.

1. Search Engine Optimization (SEO): The Theory

At its core, SEO is the science of signaling **relevance** and **authority** to a

search engine.

* **The Crawl and Index Phase:** Search engines deploy automated bots (spiders/crawlers) that traverse the web by following links. When a bot arrives at your site, it reads the HTML code and stores a snapshot of your content in a massive database called the "Index." If you are not in the index, you do not exist in search results.

* **The Ranking Phase:** When a user types a query, the search engine sorts through its index to find the most relevant answers. It uses hundreds of ranking signals to do this. Historically, this meant counting how many times a keyword appeared on a page (Keyword Density) or how many other sites linked to yours (Backlinks).

* **The Shift to E-E-A-T & GEO:** Modern SEO has evolved past simple keywords. Google's algorithm now evaluates **E-E-A-T** (Experience, Expertise, Authoritativeness, and Trustworthiness). Furthermore, with the rise of AI, we are entering the era of **Generative Engine Optimization (GEO)**. AI models (like ChatGPT or Google's AI Overviews) synthesize answers directly rather than just providing a list of blue links. To win in GEO, your content must be impeccably structured (using Schema Markup, clear heading hierarchies, and factual density) so that an AI confidently uses you as a primary citation.

2. SEO: The Practice

Translating this theory into practice requires a multi-layered approach:

* **Technical SEO:** This is your foundation. It involves ensuring your site loads in under 2 seconds (Core Web Vitals), possesses a clean `sitemap.xml` so crawlers know where to go, and uses an optimized `robots.txt` to tell bots what they are allowed to index. If the technical foundation is broken, the algorithm abandons the crawl.

* **On-Page SEO:** This is the content layer. It involves optimizing the visible text and the HTML tags. **Title Tags** and **Meta Descriptions** serve as your digital billboard—they must be highly relevant to the user's search intent. **Canonical tags** prevent the engine from getting confused if two URLs load similar content.

* **Off-Page SEO:** This represents external authority. It involves earning high-quality backlinks from other reputable sites in your industry, which acts as a "vote of confidence" in the eyes of the algorithm.

3. Google Analytics 4 (GA4): The Theory

While SEO drives traffic **to** your site, analytics helps you understand what users do once they **arrive**.

* **The Event-Driven Paradigm:** Previous analytics systems (like Universal Analytics) were "session-based." They tracked pageviews. GA4 is fundamentally different; it is an **event-based** model. Every interaction—a page load, a button click, a video play, a file download, a form submission—is recorded as an "Event." This shift allows for a much more granular understanding of the user journey, especially across different devices (mobile vs. desktop).

* **Privacy & Machine Learning:** In a world where browser cookies are disappearing due to privacy regulations (GDPR, CCPA), GA4 relies heavily on machine learning to "fill in the gaps." It uses behavioral modeling to estimate conversions when users opt out of tracking, ensuring you still have actionable data.

4. GA4 & Data Warehousing: The Practice

Implementing a modern analytics stack involves three major components:

* **The Data Collection Layer (Google Tag Manager):** Instead of writing custom JavaScript for every tracking pixel, you install a single GTM container on your site. You then use the GTM dashboard to fire specific "Tags" (like the GA4 configuration tag) based on specific "Triggers" (like a user clicking the "Start Trial" button).

* **The Reporting Layer (GA4 Dashboard):** This is where you view standard reports—where your traffic is coming from (Acquisition), what pages they view (Engagement), and whether they buy your product (Monetization).

* **The Data Warehouse Layer (BigQuery):** For enterprise systems (like `vectors.co.il``), standard dashboards are often insufficient. By exporting raw GA4 event data to **BigQuery** (Google's SQL data warehouse), you unlock the ability to run advanced analytics, combine web data with your CRM data, or feed user behavioral trends directly into your own local AI models.

Now that you understand the "why" and "what," let's dive into the "how."

1. `vectors.co.il` (Google Cloud)

Because this is a custom-hosted environment, we utilize the custom automation scripts provided in the ``src/`` directory for granular, programmatic control.

A. Generating the Sitemap

As you add new technical content or wiki articles, you need to tell search engines that the site has changed by updating the **`**Sitemap**`**.

```
```powershell
.\src\generate_sitemap.ps1 -SourceDir "C:\path\to\vectors_site\build"
```
```

This script recursively finds all HTML/MD files, extracts their modification dates, and builds a Google-compliant XML sitemap.

B. Generating an AI-Optimized Robots.txt

To ensure that Generative AI bots (GPTBot, ClaudeBot, etc.) are allowed to ingest your site and cite you as a source, you must define permissions in your **`**Robots.txt**`**.

```
```powershell
.\src\generate_robots.ps1 -OutputDir "C:\path\to\vectors_site\build" -SitemapUrl
"https://vectors.co.il/sitemap.xml"
```
```

C. Injecting SEO and GA4 Tags

Instead of manually editing hundreds of HTML files, use this script to automatically inject **`**Title Tags**`**, **`**Meta Descriptions**`**, **`**Canonical Tags**`**, **`**HREFLANG**`** localization, and the **`**Google Tag Manager (GTM)**`** code directly into your static files.

```
```powershell
.\src\inject_full_seo_tags.ps1 -TargetDir "C:\path\to\vectors_site\build" -GtmId
"GTM-XXXXXXX"
```



```

Note: Replace `GTM-XXXXXXX` with your actual Google Tag Manager ID. The Tag Manager will then handle forwarding data to **GA4 and eventually to **BigQuery**.**

D. Monitoring AI Mentions

To see how AI models categorize your platform, you can use your local **LLM** setup. This script searches for mentions of your brand and asks the AI to analyze the context.

```
```powershell
python src/ai_mentions_monitor.py --brand "vectors.co.il" --model "llama3"
```
```

2. ginsbourg.co.il (Wix)

Because Wix is a **Managed CMS**, file-level scripting is not applicable. All integrations must be performed via the Wix Dashboard.

A. SEO Configuration

1. **Initial Setup:** Navigate to **Marketing & SEO** > **SEO Setup Checklist**. This built-in tool natively connects your site to **Google Search Console (GSC)** so Google can start indexing it.
2. **Schema Markup:** Go to **SEO Settings** > **Advanced** > **Structured Data** to input your Service and Organization **JSON-LD**. This will help you get rich results like service snippets in search.
3. **AI Visibility:** Use the Wix AI assistant to monitor and standardize your professional bio across LLMs.

B. Google Analytics 4 Integration

1. Do not manually paste GA4 script codes into the site's custom code section. This can cause duplicate tracking.
2. Navigate to **Marketing & SEO** > **Marketing Integrations**.

3. Select **Google Analytics** and paste your Measurement ID (`G-XXXXXXX`). Wix natively handles the rest.
4. **Enhanced Tracking:** Ensure "Form Interactions" is enabled directly inside your Google Analytics admin panel to automatically track your "Contact Me" inquiries without writing custom event code.

🌐 Future Roadmap

- * **Competitor Analysis:** Compare your GEO score against top competitors.
- * **Multimodal Analysis:** Deep scan of images and alt text for accessibility and AI vision understanding.
- * **Agentic Protocol (MCP) Support:** Validation for agent-specific endpoints.

Built for the future of search.